Summary in Graph

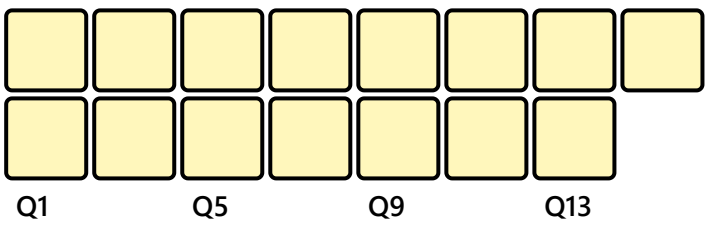
Exam Summary (GO Classes GATE CS | Computer Networks | Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (61) out of 608 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Computer Networks

An organisation has been assigned the Class A network $12.0.0.0$. The network administrator wants to divide this network into 32 equal-sized subnets.
What would be the subnet mask of the subnet to which the IP address $12.45.78.10$ belongs?

A. $255.255.0.0$

B. $255.248.0.0$
C. $255.255.248.0$
D. $255.254.0.0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

An organisation has been allocated the IP address block $140.25.0.0$. The goal is to divide this address block into multiple subnets, each capable of supporting at least 25 hosts.
Which of the following subnet masks should be used to create subnets that meet this requirement while minimising address space wastage?

A. $255.255.255.192(/26)$
B. $255.255.255.128(/25)$
C. $255.255.255.224(/27)$
D. $255.255.255.240(/28)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Assume an organisation is assigned the IP address block $140.25.0.0/16$. The organisation decides to subnet this block using a $/27$ mask.
What is the maximum number of such $/27$ subnets that can be formed from the original address block?

A. 256
B. 512
C. 8192
D. 2048

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A subnet is created from the network $140.25.0.0$ using the subnet mask $255.255.255.224$ (i.e., /27).
If the subnet is required to support 25 hosts, how many IP addresses within that subnet will not be utilised (including both unused and reserved addresses)?

A. 2
B. 5
C. 7
D. 1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 02sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A subnet is created from the IP address block $165.65.y.z/16$ using a subnet mask of $255.255.240.0(/20)$.
If the 3rd subnet (subnet ID 0011) is being used, what is the first assignable IPv4 address in this subnet?

A. $165.65.48.0$
B. $165.65.48.1$
C. $165.65.48.254$
D. $165.65.49.1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A company is assigned three IP address blocks: $149.16.23.0/22$, $149.16.24.0/22$, and $149.16.30.0/22$. It wants to divide these blocks into 12 equal-sized subnets for internal use. What subnet mask should be assigned to each of the 12 subnets?

A. $/22$
B. $/23$
C. $/24$
D. $/25$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A company receives a $/22$ IP address block and wants to divide it into equal-sized subnets for internal use. How many $/24$ subnets can be created from a single $/22$ block?
A. 2
B. 4
C. 8
D 16

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

You are asked to allocate a subnet from the block $214.97.254.0/17$ that can provide exactly 2 usable host IP addresses.
Which of the following subnet assignments is most suitable?

A. $214.97.254.4/30$
B. $214.97.254.0/29$
C. $214.97.254.2/31$
D. $214.97.254.0/28$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A provider has been assigned the IP block **128.30.0.0/24**.
It needs to allocate subnets to three customers with the following host requirements:

- **Customer C**: 80 hosts
- **Customer B**: 60 hosts

- **Customer A**: 40 hosts

What is the most efficient order in which the subnets should be allocated to minimize address wastage?

A. A → B → C
B. C → B → A
C. B → A → C
D. Any order will result in the same usage

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

A company has been assigned a Class C IP address and decides to use 6 bits for subnetting.
How many usable host addresses per subnet are available?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

From the address block $140.25.0.0$, a /27 subnet is created, which contains 32 IP addresses. If only 25 of these addresses are used or assigned, what percentage of the subnet's address space is not utilised?

A. 12.5%
B. 21.875
C. 6.25%
D. 25%

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

An organization has subnetted the IP block $200.1.1.0/24$ using Variable Length Subnet Masking (VLSM) as follows:

Host Requirements:

Subnet A: 72 hosts
Subnet B: 35 hosts
Subnet C: 20 hosts
Subnet D: 18 hosts

```
                 200.1.1.0/24
                     |
      +-------------+-------------+
      |                           |
200.1.1.0/25                200.1.1.128/25
  [Subnet A]                    |
                     +----------+-----------+
                     |                      |
             200.1.1.128/26        200.1.1.192/26
              [Subnet B]               |
                              +--------+--------+
                              |                 |
                     200.1.1.192/27     200.1.1.224/27
                      [Subnet C]         [Subnet D]
```

The organization now needs to add a **fifth subnet (Subnet E),** requiring **20 new hosts**. Which of the following is the most appropriate way to accommodate Subnet E *without requesting a new network block*?

A. Convert Subnet D from `/27` to `/28` and use the leftover space for Subnet E
B. Merge Subnets B and C to free up a `/25` block for subnet E.
C. Split Subnet A (`200.1.1.0/25`) into four `/27` subnets, assign three to Subnet A and one to Subnet E
D. Use the unused space after Subnet D (`200.1.1.255/32`) to create Subnet E

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A company uses part of the $214.97.254.0/17$ block.
The subnet B is allocated $214.97.254.0/25$, and $214.97.254.0/29$ is reserved for internal routing.
How many usable addresses are left in Subnet B for host assignments?

A. 128
B. 121
C. 120
D. 119

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

The provider has allocated the block $128.30.0.0/24$ and assigned the first 128 IPs to a customer needing 80 hosts. The next customer requires 60 hosts.
Which subnet should be assigned to this second customer?

A. $128.30.0.192/26$
B. $128.30.0.64/26$
C. $128.30.0.128/26$
D. $128.30.0.128/27$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

An organisation has been allocated the IP block $10.0.0.0/24$ to set up a subnet LAN1 that must support 40 hosts, including space for a network address, a broadcast address, and one router interface.
Two subnetting strategies are considered:

- Method 1: **Use a single subnet of size /26, which allocates 64 IP addresses.**
- Method 2: **Split LAN1 into two smaller subnets**: one subnet with ***32 addresses (/27)*** and one subnet with ***16 addresses (/28)***

Which of the following statements is correct regarding address utilisation?

A. Method 1 is more efficient because it uses a single continuous block of 64 addresses.

B. Method 2 is invalid since subnets of this size $/27$ and $/28$ cannot coexist in the same /24 block.

C. Both methods are equally efficient since both satisfy the 40-host requirement.

D. Method 2 is more efficient, as it allocates only $32 + 16 = 48$ addresses instead of 64.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff